Week 2 Assignment: Data Import, Export, and Exploratory Data Analysis (EDA)

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Course: ANA 505 Foundations of Data Mining

1. Dataset Preparation

The dataset employee_data_week2.csv containing 100 employee records and 6 variables was successfully loaded and structured.

Data Structure (Variables & Types):

Variable	Data Type	Classification
EmployeeID	int64 (Numeric)	Identifier
Age	int64 (Numeric)	Numeric
Department	character (String)	Categorical
Salary	int64 (Numeric)	Numeric
Experience	float64 (Numeric)	Numeric
Education_Level	character (String)	Categorical

2. Descriptive Summary

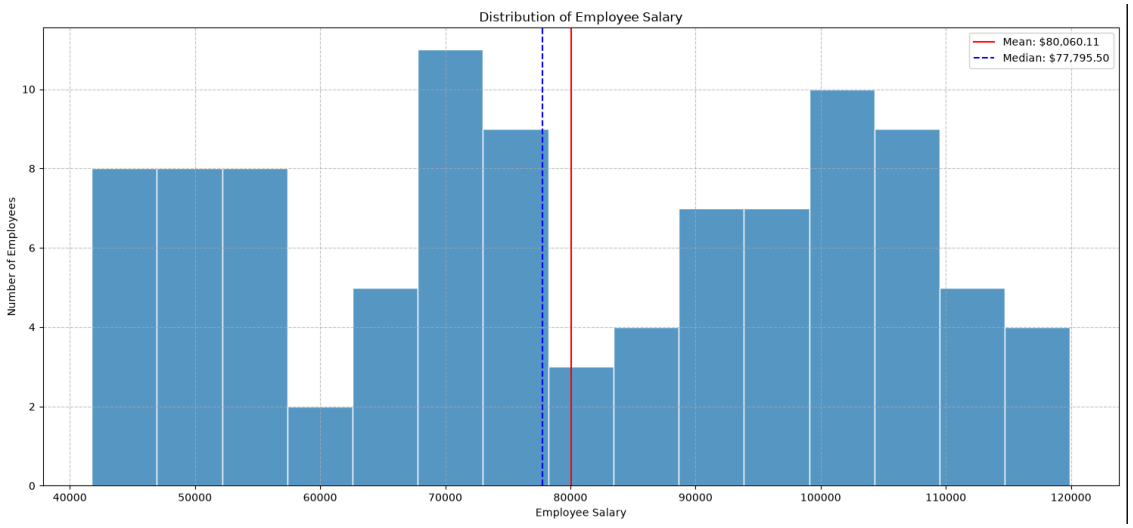
There are no missing values across the 100 records in the dataset.

Summary Statistics (Numerical):

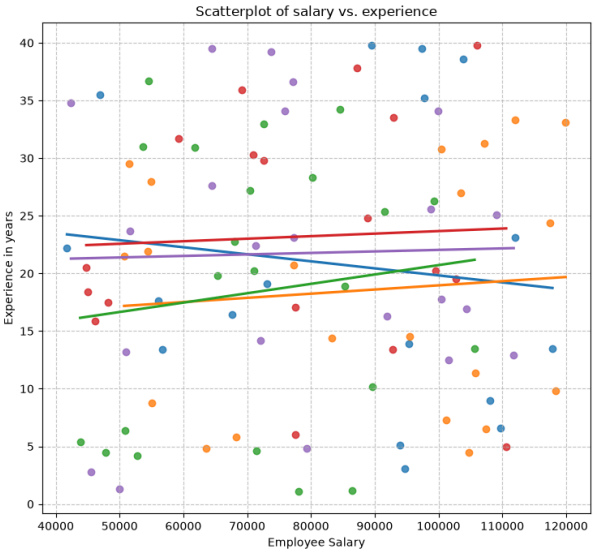
Statistic	Salary	Experience
Min	$41,754.00	1.10
Mean	$80,060.11	20.42
Median	$77,795.50	20.20
Max	$119,921.00	39.80
Variance	503,382,388.12	128.12
Std Dev	$22,436.18	11.32
Range	$78,167.00	38.70

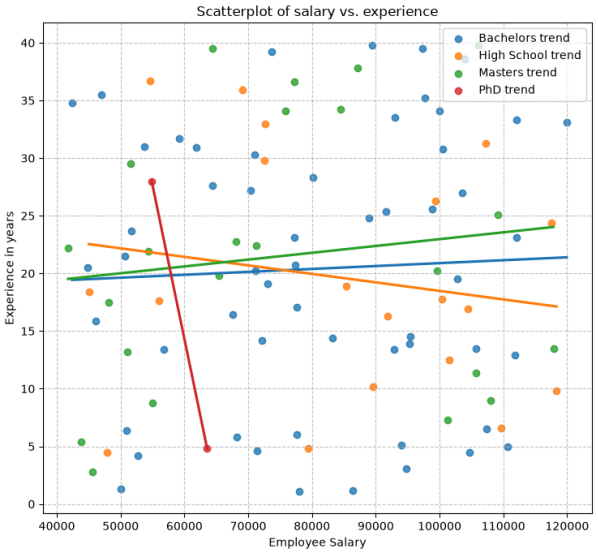
3. Basic Data Visualization

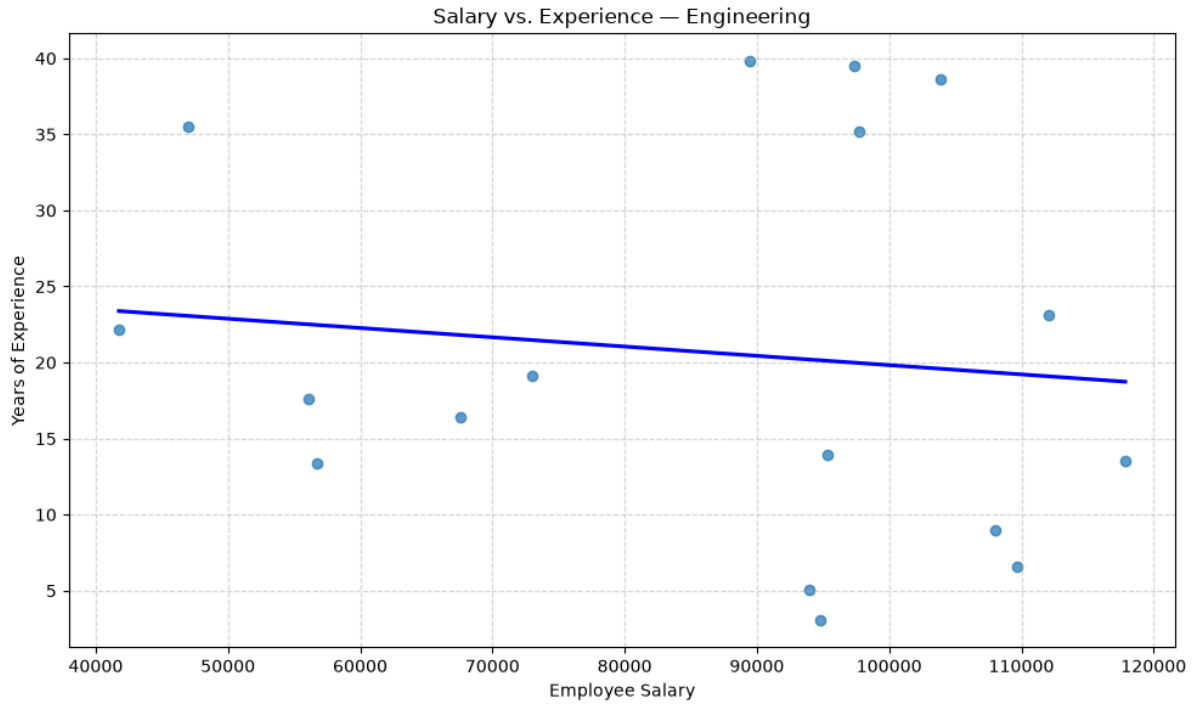
The visualizations are generated via the execution of the included Python and R scripts.




Scatterplot of salary vs. experience
Experience in years
Employee Salary
0
5
10
15
20
25
30
35
40
40000
50000
60000
70000
80000
90000
100000
110000
120000


Scatterplot of salary vs. experience
Experience in years
Employee Salary
0
5
10
15
20
25
30
35
40
40000
50000
60000
70000
80000
90000
100000
110000
120000


Scatterplot of salary vs. experience
Bachelors trend
High School trend
Masters trend
PhD trend
Experience in years
Employee Salary
0
5
10
15
20
25
30
35
40
40000
50000
60000
70000
80000
90000
100000
110000
120000



4. Reflection

The exploratory data analysis of the employee_data_week2.csv dataset yielded several insights into the relationships between employee experience, department, education level, and salary. The dataset comprises 100 complete records with no missing values. The summary statistics indicate that the average salary is roughly $80,060, while the average experience level is around 20.4 years. The distribution of salaries, as visualized in the histogram, shows a spread from around $41,754 up to $119,921, with the median ($77,795) closely tracking the mean, suggesting a relatively symmetric distribution without extreme skewness though the histogram reveals it appears somewhat uniform rather than following a perfect normal curve.

The visualizations significantly enhanced my understanding of these underlying trends. While summary statistics provide a numerical baseline, the scatterplot clearly displays a positive correlation overall as years of experience increase, salary tends to increase as well. However, the wide variance in compensation at similar experience levels implies that experience alone does not fully explain salary differences, highlighting the need for future multivariate models to examine other factors like department and education.

It's a basic activity and a good refresher working on visualization and exploring a dataset, but to understand the data further, it's worth investigating other variables that could explain the salary trends like the education level effect on salary or department vs. salary

5. Appendix: R Script

Python

```
# Install required packages if they are not already installed
required_packages <- c("ggplot2", "dplyr")
missing_packages <- required_packages[
  !sapply(X = required_packages, FUN = requireNamespace, quietly = TRUE)
]
if (length(missing_packages) > 0) {
  install.packages(missing_packages, dependencies = TRUE)
}
library(ggplot2)
library(dplyr)
#============================================================
## 1. Read the dataset
#============================================================
fname <- "../datasets/employee_data_week2.csv"
if (!file.exists(fname)) {
  stop(
    "Dataset not found. Make sure employee_data_week2.csv ",
    "is in the ../datasets/ folder."
  )
}
employee_data <- read.csv(fname, stringsAsFactors = FALSE)
# Display the first few rows
head(employee_data)
# Display the dataset structure
str(employee_data)
#============================================================
## 2. Data exploration
#============================================================
# Summary statistics for all variables
summary(employee_data)
# Summary statistics for Salary and Experience
required_variables <- c("Salary", "Experience")
summary(employee_data[required_variables])
```

```r
# Calculate requested statistics
stats <- data.frame(
  Variable = required_variables,
  Mean = sapply(
    employee_data[required_variables],
    mean,
    na.rm = TRUE
  ),
  Median = sapply(
    employee_data[required_variables],
    median,
    na.rm = TRUE
  ),
  Variance = sapply(
    employee_data[required_variables],
    var,
    na.rm = TRUE
  ),
  Standard_Deviation = sapply(
    employee_data[required_variables],
    sd,
    na.rm = TRUE
  ),
  Minimum = sapply(
    employee_data[required_variables],
    min,
    na.rm = TRUE
  ),
  Maximum = sapply(
    employee_data[required_variables],
    max,
    na.rm = TRUE
  ),
  Range = sapply(
    employee_data[required_variables],
    function(x) diff(range(x, na.rm = TRUE))
  )
)
stats
# Check for missing values
colSums(is.na(employee_data))
#============================================================
## 3. Visualizations
#============================================================
```

```r
## 3.1 Histogram: Salary
# Calculate mean and median salary
mean_salary <- mean(employee_data$Salary, na.rm = TRUE)
median_salary <- median(employee_data$Salary, na.rm = TRUE)
# Create salary histogram
ggplot(employee_data, aes(x = Salary)) +
  geom_histogram(
    bins = 15,
    color = "white",
    fill = "steelblue"
  ) +
  geom_vline(
    xintercept = mean_salary,
    color = "red",
    linetype = "solid"
  ) +
  geom_vline(
    xintercept = median_salary,
    color = "blue",
    linetype = "dashed"
  ) +
  labs(
    title = "Distribution of Employee Salary",
    x = "Employee Salary",
    y = "Number of Employees"
  ) +
  theme_minimal()
## 3.2 Scatterplot: Salary and Experience by department
ggplot(
  employee_data,
  aes(x = Salary, y = Experience, color = Department)
) +
  geom_point() +
  geom_smooth(
    method = "lm",
    se = FALSE
  ) +
  labs(
    title = "Salary vs. Experience by Department",
    x = "Employee Salary",
    y = "Experience in Years",
    color = "Department"
  ) +
  theme_minimal()
```

```r
## 3.3 Scatterplot: Salary and Experience by education level
ggplot(
  employee_data,
  aes(x = Salary, y = Experience, color = Education_Level)
) +
  geom_point() +
  geom_smooth(
    method = "lm",
    se = FALSE
  ) +
  labs(
    title = "Salary vs. Experience by Education Level",
    x = "Employee Salary",
    y = "Experience in Years",
    color = "Education Level"
  ) +
  theme_minimal()
## 3.4 Scatterplot: Engineering employees
engineering_data <- employee_data %>%
  filter(Department == "Engineering")
ggplot(
  engineering_data,
  aes(x = Salary, y = Experience)
) +
  geom_point(color = "steelblue") +
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "red"
  ) +
  labs(
    title = "Salary vs. Experience - Engineering",
    x = "Employee Salary",
    y = "Experience in Years"
  ) +
  theme_minimal()
#============================================================
## 4. Data export
#============================================================
# Create three salary categories using tertile cutoffs,
# matching the Python qcut(..., q = 3) code.
salary_cutoffs <- quantile(
  employee_data$Salary,
  probs = c(1 / 3, 2 / 3),
```

```r
  na.rm = TRUE
)
employee_data$Salary_Category <- cut(
  employee_data$Salary,
  breaks = c(-Inf, salary_cutoffs, Inf),
  labels = c("Low", "Medium", "High"),
  include.lowest = TRUE
)
# Review the salary categories
table(employee_data$Salary_Category)
# Save the modified dataset
modified_path <- "../datasets/employee_data_week2_modified.csv"
write.csv(
  employee_data,
  file = modified_path,
  row.names = FALSE
)
cat("Modified dataset saved to:", modified_path, "\n")
```